GONCA ARABACI

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EXPERIENCE

Computer Engineering Intern | Hifive Games, İstanbul *Feb 2026 – Jun 2026*

- Built modular, event-driven software architecture in C# using EventBus messaging and the ServiceLocator pattern to improve maintainability and decoupling.
- Improved runtime performance through object pooling, lifecycle management, and zero-allocation techniques.

Computer Engineering Intern | Eps Teknoloji, İzmir *Jul 2024 – Aug 2024*

- Developed an industrial computer vision pipeline with OpenCV and Pypylon for automated fabric defect detection and dimensional measurement, improving optical sensor accuracy.
- Implemented advanced shape and contour detection algorithms; configured secure remote access to Raspberry Pi–based vision systems via NAT gateway.

Computer Engineering Intern | Enm Digital, Manisa *Mar 2024 – Jun 2024*

- Contributed to frontend development for AI-driven web applications, integrating AI features to improve UX and applying rigorous testing practices.

Computer Engineering Intern | Codeo, İzmir *Aug 2023 – Sep 2023*

- Built grid- and lock-based gameplay logic in Unity, applying SOLID principles for maintainable, extensible code.

EDUCATION

B.Sc. in Computer Engineering — Manisa Celal Bayar University, Manisa *Sep 2021 – Jun 2026*
GPA: 3.39 / 4.0

PROJECTS

Stratis — Graduation Project 1 *Oct 2025 - Jan 2026*

Built a tactical AI simulation in Unity using ML-Agents (PPO), combining reinforcement learning with rule-based decision logic for real-time combat behavior.

TUSAŞ Lift Up — Graduation Project 2 *Oct 2025 - Jun 2026*
Developed a DIC-based application analyzing strain concentration and displacement-field evolution as fatigue crack indicators; validated accuracy against experimental fatigue test data.

Teknofest Air Defense System *Feb 2025 - Jun 2025*
Led the software team building a real-time embedded CV system detecting shapes and colors up to 10m; developed detection algorithms from scratch and optimized them for embedded hardware.

Global Game Jam 2026 : Mask Heist *Jan 2026*
Implemented network-based character synchronization and real-time data management for a multiplayer game built under a 48-hour deadline.

TanIA — NLP Healthcare Decision Support *Jul 2024*
Contributed to an AI-powered system analyzing patient symptoms, generating preliminary assessments, and routing to the appropriate medical specialty.

TUBİTAK 2209-A: Development of a Solar Panel that Changes its Position Based on the Analysis of Meteorological Data and Examination of Energy Efficiency *Mar 2024 - Apr 2025*
I developed a solar panel system that adjusts its orientation based on meteorological data and studied energy efficiency through dynamic positioning.

TECHNICAL SKILLS

Languages: Python, C#, Java
Computer Vision: OpenCV, Pypylon, Digital Image Correlation (DIC), Contour/Shape/Object Detection
Software Engineering: OOP, SOLID Principles, Design Patterns, Data Structures & Algorithms, Agile/Scrum
Tools & Platforms: Git, Raspberry Pi, Unity, Visual Studio, VS Code

LEADERSHIP

Co-Lead, Oyun Tasarımı ve Geliştiricileri Topluluğu *Sep 2024 – Jan 2026*
Organized 20+ events and 3 game jams, managing GSB-funded projects while leading creative direction.

Co-Lead, Dijital Oyun Geliştirme Ekibi Nov 2023 – Sep 2024
Coordinated a community focused on game development events.